

Full Approximation of Deep Neural Networks through Efficient Optimization

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Abstract—Approximate Computing is a promising paradigm for mitigating computational requirements of Deep Neural Networks (DNN), by taking advantage of their inherent error resilience. Specifically, the use of approximate multipliers in DNN inference can lead to significant improvements in power consumption of embedded DNN applications. This paper presents a methodology for efficient approximate multiplier selection and for full and uniform approximation of large DNNs, through retraining and minimization of the approximation error. We evaluate our methodology using 422 approximate multipliers from the EvoApprox library, with three different Residual architectures trained with Cifar10, and achieve energy savings of up to 18% surpassing the original floating-point accuracy, and of up to 58% with an accuracy loss of 0.73%.

I. INTRODUCTION

Through approximate computing, hardware requirements in error-resilient applications such as DNNs can be reduced. Approximations at software level are e.g. precision scaling or quantization [1], [2], [3]. At hardware level, approximation of arithmetic units such as multipliers [4] can lead to significant energy savings. In this context, DNNs can be partially or fully approximated. Techniques for partial approximation include retraining [4], [5], and more recently, authors in [6] proposed *ALWANN*, a framework for partial DNN approximation without retraining. Drawbacks of partial approximation are the energy savings, limited by the amount of approximated elements, and its challenging implementation in generic hardware accelerators. Furthermore, the search for optimal designs is highly time-consuming in traditional simulation frameworks [6]. On the other hand, through specialized methodologies for full approximation of DNNs, a more generic implementation and more energy savings can be achieved. In [7], authors investigate fully approximated and retrained DNNs with quantized weights and FP activations. Methods for optimizing the Approximate Multiplier (AM) itself were introduced in [8], [9], which proved to be effective but time-consuming and with low flexibility, and authors in [10], [11] demonstrated the viability of accuracy recovery through hardware-aware retraining in DNNs for digit recognition. Specialized simulation frameworks such as *Concrete* [12], based on Caffe [13], have been recently proposed to further accelerate approximate DNN computation in Graphic Processing Units (GPUs) with small time overhead, making the optimization of larger approximate DNNs more attractive.

Considering all currently available approximate hardware designs, selecting the appropriate AM for uniform DNN inference without accuracy loss is a non trivial task, which depends mainly on the availability of resources for DNN retraining. For some applications, such as those with proprietary DNNs, retraining with the original dataset is not possible as it is not available to the public, and therefore, other approaches to overcome this are needed. On the other hand, retraining with all possible design points given all existing AMs is a time exhaustive task even with specialized simulation frameworks. For example, we would need around 1.5 days just to train a ResNet20 [14] with every multiplier from [15] for only one epoch with our specialized simulation framework. With this motivation, we present a methodology for efficient multiplier selection and full approximation of large DNNs without accuracy loss. We focus on designs with the same multiplier for all convolutional and fully-connected (FC) layers, which is desirable for hardware accelerators with uniform processing elements. For this, given an initial set of AMs, we first select the Pareto set of power consumption and Mean Relative Error (MRE), as we prove that MRE is highly correlated to DNN accuracy. Through this, we restrict our search to 18% of the initial design space. Then, we retrain only the approximate DNNs with multipliers from this Pareto set. Our methodology allows retraining even without the original dataset, as we can switch to minimizing the propagated approximation error instead of optimizing the DNN accuracy itself.

The novel contributions presented in this work are:

- A data-driven analysis for multiplier selection based on MRE, focused on convolutional and FC layers.
- a methodology for efficient uniform and full approximation of DNNs.
- An alternative to DNN retraining when the original training dataset is not available.
- Significant power improvements without accuracy loss of three residual architectures [14] trained with Cifar10, through simulation of 422 AMs from the EvoApprox library [15].

II. PRELIMINARIES

A. Deep Neural Networks

DNNs are characterized by the use of convolutional and FC layers. These operations can be generalized as follows: $y =$